

Disperse to ignite, concentrate to endure: a decay-driven crossover in modular complex contagions

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Abstract. Should a costly new practice be seeded into a few dense teams, or scattered across the organization? We study this on synthetic two-layer organizations of dense small teams, using fractional-threshold contagion (heterogeneous thresholds, innovator atom, credibility-weighted exposure) under a paired protocol: within each condition, every strategy runs on the same 50 organizations with identical agent draws. Without decay, dispersed seeding dominates the tested ignition band: +39.7 percentage points of terminal adoption over cluster seeding at the headline setting (95% CI [35.4, 43.9]), winning 14 of 40 cells of a threshold–budget map (23 saturated or starved cells equivalent within ± 2 pp, 3 uncertain, no cluster win of ≥ 2 pp). Under reinforcement-dependent decay the terminal-adoption advantage shrinks and reverses between the tested relapse rates: dispersed cascades are increasingly throttled, while clustered adoption stays near 30% across the tested decay settings. The sign of the contrast replicates on a real modular topology (email-Eu-core, synthetic behavioral attributes). All diffusion outcomes are simulated and all behavioral attributes synthetic; version-controlled code with frozen seeds reproduces every number.

Keywords: complex contagion, threshold models, seeding strategies, organizational networks, agent-based simulation

1 Introduction and related work

Threshold models [4] make costly adoption require reinforcement from multiple credible sources. Cascades on random graphs obey a vulnerability boundary [13]; complex contagions need wide bridges of overlapping ties, which long-range shortcuts destroy [2, 1]. Network-intervention taxonomies [11] stress matching the intervention to context; the expectation that clustered seeding should win follows from those wide-bridges results—evidenced largely in populations not already embedded in cohesive groups. Adjacent results strain it: modularity aids cascades only up to an optimum [7]; dispersed neighbor seeding beats clustered seeding in threshold models [8]; seed placement is its own measurement problem [5, 10]; random ties match or beat clustered ties unless interest decays fast [12, 9]—a structural reach–reinforcement trade-off whose seeding-strategy analogue we test at fixed topology, asking when decay reverses the ordering.

Real organizations are modular—nearly everyone already sits inside a dense small team—so what does cluster seeding buy when every seed is already surrounded by a cluster? We answer with a transparent agent-based model under a paired protocol pre-specified in a version-controlled decision log.

2 Model and paired protocol

A generator produces two-layer organizations: a formal hierarchy (8 departments, 249 line teams of mean size 8, $N=2000$) and an informal influence network—dense within teams ($p=0.9$), locally concentrated between ring-adjacent “sister” teams, sparse across departments (silo 0.85)—on which adoption propagates. Agent i holds threshold $\theta_i \sim \text{Beta}(\mu\kappa, (1-\mu)\kappa)$, $\mu=0.30$, $\kappa=20$, and is an innovator ($\theta_i=0$) with probability 0.025. Exposure is the credibility-weighted share of contacts whose adoption is visible,

$$\text{share}_i(t) = \frac{\sum_{j \in N(i)} v_j w_{ij} a_j(t-1) + b(t)}{\sum_{j \in N(i)} w_{ij} + b(t)}, \quad (1)$$

with credibility w (close collaborator 1.0 > manager 0.7 > distant peer 0.6 > comms 0.3), source visibility v_j (default 1), and $b(t)$ a transient comms term during a broadcast. Adoption requires ready ($\text{share}_i \geq \theta_i$ and $\text{share}_i > 0$), willing (Bernoulli 0.85), and able (1.0 here) [3]; updates are synchronous. Decay (off unless stated): an adopter relapses with probability ρ per step while $\text{share}_i < r\theta_i$, where r sets how much reinforcement retention requires. We report terminal adoption (adopted share among active agents at the fixed point; 98–100% converge) and, secondarily, cumulative reach (ever-adopted share). Four seeded strategies share a 5% budget (100 seeds): random, champions (top informal degree), cluster (whole teams), one-per-team (round-robin coverage); broadcast is a zero-seed awareness baseline—fixed heuristics, not influence-maximization optima [6, 8].

Paired protocol. Within each condition cell, every strategy runs on the same 50 fresh organizations with identical agent draws (common random numbers; cells may use independent samples), so contrasts are within-organization paired differences—confound-free at the organization level, with no efficiency gain over independent sampling assumed. Two confirmatory families (regime map, decay grid) were pre-specified in a version-controlled decision log—after exploratory $n=12$ panels, before the confirmatory run—and Holm-corrected for both difference and TOST-equivalence against a ± 2 pp practical band; each cell gets a three-way verdict: win, equivalent (within ± 2 pp, for the tested cells and model), or uncertain. Seeds were frozen before execution; all paired results come from a single successful full-pipeline execution (6350 simulations).

3 Results

Without decay, dispersion dominates the ignition regime. At the headline setting, random seeding reaches $72.0 \pm 9.2\%$ terminal adoption (mean $\pm$ sd) against $32.4 \pm$

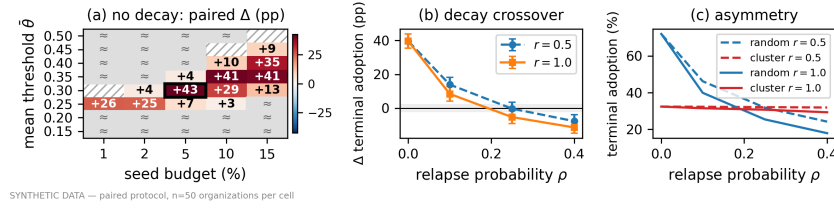


Fig. 1. Synthetic data. (a) Paired Δ terminal adoption (random – cluster, pp) over $\bar{\theta} \times$ seed budget ($n=50$ pairs/cell; Holm-corrected difference and TOST families): red = dispersion wins; “ $\approx$ ” = equivalent within ± 2 pp; hatched = uncertain; box = headline setting. (b) The same contrast under decay (95% CIs; ± 2 pp band). (c) Absolute levels: clustered adoption barely moves; dispersed collapses.

11.0% for cluster seeding: a paired difference of **+39.7 pp**, 95% CI [35.4, 43.9]. Degree targeting adds ~ 3.9 pp (descriptive), and coverage with one seed per team yields no clear difference from random (-1.4 pp, CI $[-5.0, +2.2]$)—on average, random already lands seeds in 82 of 249 line teams (~ 205 end at majority) vs. 13 (~ 91) for cluster. Across the map (Fig. 1a), dispersion wins 14 of 40 cells—the diagonal ignition band, $+2.8$ to $+42.6$ pp ([39.1, 46.0] at the boxed cell, an independent paired sample)—the strategies are equivalent within ± 2 pp (TOST) in 23 cells (saturation at low $\bar{\theta}$, starvation at high $\bar{\theta}$ /low budget), 3 stay uncertain; no cell shows a practically relevant cluster win (≥ 2 pp)—the largest cluster edge, 0.8 pp in the starved corner, sits inside the band. Broadcast converts $5.5 \pm 3.4\%$ (0.0% in all 12 runs without innovators; independent panel).

Decay drives a crossover. With relapse active (Fig. 1b,c), the dispersion advantage falls monotonically in ρ and flips sign between the tested values. When retention requires full reinforcement ($r=1$): $+8.4$ pp [4.2, 12.7] at $\rho=0.10$, -5.3 pp $[-9.0, -1.5]$ at $\rho=0.25$, -11.4 pp $[-14.8, -8.0]$ at $\rho=0.40$ —the crossover lies between $\rho=0.10$ and 0.25 . At the weaker coupling $r=0.5$ it comes later: $+13.9$ pp [9.6, 18.3] at $\rho=0.10$, parity still uncertain at $\rho=0.25$ (-0.3 pp $[-4.1, +3.5]$), reversal at $\rho=0.40$ (-7.7 pp $[-11.5, -3.9]$). The crossover is asymmetric: clustered adoption barely moves (32.4% at $\rho=0$ to 29.3% at the harshest setting)—it endures rather than wins—while dispersed adoption collapses (72.0% to 17.9%). It is also endpoint-dependent: at ($r=1$, $\rho=0.25$) the terminal contrast has reversed, but the cumulative-reach contrast (secondary, descriptive) is still uncertain (-1.8 pp $[-5.4, +1.8]$; clearly negative only at $\rho=0.40$: -7.4 pp $[-10.7, -4.1]$)—so the terminal reversal there is driven mainly by differential retention, 0.99 (cluster) vs. 0.86 (random). Relapse throttles the dispersed cascade while it grows: random’s ever-adopted share is 29.0% vs. 72% without decay.

Robustness and a real topology. The no-decay contrast keeps its sign across one-at-a-time organization variants ($N \in \{500, 2000, 8000\}$, team size $\{5, 12\}$, silo $\{0.5, 0.7, 0.95\}$; descriptive): paired Δ from $+14.7$ to $+48.3$ pp, all CIs excluding zero. On the real email-Eu-core topology (986 nodes, 42 ground-truth departments; all behavioral attributes synthetic) the contrast keeps its sign: $+13.5$ pp [1.0, 26.0] over 50 paired attribute draws. Adoption there is bimodal: draws end low (median 9%) or high (median 85%), none between 21% and 82%. This is a

sign replication on a real modular topology, not an empirical validation of the mechanism.

4 Discussion

Once everyone already lives in a dense team, guaranteed local reinforcement—cluster seeding’s defining asset—is no longer scarce; scattering buys more ignition surface per seed, and decay then decides whether dispersed ignitions compound into organization-wide growth: relapse throttles the growing cascade, while clustered adoption sits on local critical mass relapse barely touches. The prescription is conditional and asymmetric: disperse to ignite when adoption reinforces itself; concentrate to endure when reinforcement erodes quickly. A local-ignition predictor, pre-specified in the same log, failed its validation gates (AUC 0.64, below even the 0.70 partial threshold): most dispersed-seeding ignition is not explained by the local predictor; a two-level model is future work. All of this holds within a synthetic, uncalibrated model family—thresholds unobserved, time abstract, decay parameters stipulated on a grid; a 21-entry limitations file ships with the code.

Reproducibility. Every modeling decision (D1–D23) is logged with alternatives; analysis families pre-specified there; seeds frozen; one successful full-pipeline execution for the paired results. Every statistic is machine-extracted from the versioned results and checked by an audit script. Repository (AGPL-3.0): github.com/PaulBufort/adoption-sim.

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